

Answer: B

- 31** When an electron approaches a uniform magnetic field at an angle, it will proceed to move in a helical path.

$$F = Bqv \sin \theta$$

$$7.3 \times 10^{-16} = (0.084)(1.6 \times 10^{-19})v \sin 23^\circ$$

$$v = 1.4 \times 10^5 \text{ m s}^{-1}$$

Answer: A

- 32** Using Faraday's Law of Electromagnetic Induction